C_3 Pathways VS C_4 Pathways

C_3 Pathways		C_4 Pathways
RuBP (Ribulose-1,5-bisphosphate) **(5C)**	**PRIMARY CO_2 ACCEPTOR**	**PEP** (Phosphoenol pyruvate) **(3C)**
3-Phosphoglycerate (PGA)	**FIRST STABLE PRODUCT**	Oxaloacetic acid (OAA)
Mesophyll cells	**SITE OF CO_2 FIXATION**	Initial fixation: Mesophyll cells Calvin cycle: Bundle sheath cells
One (Mesophyll)	**CELL TYPES INVOLVED IN CO_2 FIXATION**	Two (Mesophyll + Bundle sheath)
Absent	**PRESENCE OF KRANZ ANATOMY**	Present
Mesophyll cells	**RUBISCO LOCATION**	Bundle sheath cells only

Absent	**PEP CARBOXYLASE**	Present in mesophyll cells
High	**PHOTORESPIRATION**	Negligible
Low under high temperature / low CO_2	**CO_2 FIXATION EFFICIENCY**	Effective even in strong sunlight & low CO_2
3 ATP consumed per CO_2	**ATP CONSUMPTION**	5 ATP total (3 in Calvin cycle + 2 extra for C_4 pump)
High	**PHOTORESPIRATION AT LOW LIGHT:**	Negligible
High	**PHOTORESPIRATION AT HIGH LIGHT:**	Negligible
High photorespiration	**RESPONSE TO LOW CO_2**	Negligible photorespiration

Photorespiration decreases	**RESPONSE TO HIGH CO_2**	Always negligible
20–25°C	**TEMPERATURE OPTIMUM**	30–40°C
Wheat, rice, barley, spinach	**EXAMPLES**	Sugarcane, maize, sorghum, millet, Amaranthus